Gen-AI in assessment declaration

Student 500 number:_____500786680_____

Module Code:_____4711_____

Assessment:_____

Assessment's Gen-AI Category (see the Assessment Brief): 2 or 3

The following statement reflects my use of Gen-AI within this assessment (select all that apply):

Description of use	**Select (X)**
Gen-AI has not been used at all in producing this assessment.	
Gen-AI was used in an assistive role to research topics and provide explanations.	
Gen-AI was used in an assistive role to suggest sources of information, e.g. publications to read.	
Gen-AI was used in an assistive role to summarize the content of publications or other sources.	
Gen-AI was used in an assistive role to produce an outline or draft structure of my assessment.	

Note: for Category 2 assessments, the fundamental content of the submission (e.g., the arguments, interpretations, conclusions etc) must be your own.	
Gen-AI was used in an assistive role to improve spelling, grammar, sentence structure and/or clarity of expression of my assessment. Note: for Category 2 assessments, the basic written expression must be your own. So Therefore whilst Gen-AI may correct mistakes it must not produce the actual content.	
Gen-AI was used in an assistive role to translate text during the production of the work.	
Gen-AI was used in an assistive role to analyse data and/or produce visualizations of data.	
Gen-AI was used in an assistive role to help produce creative outputs e.g. art, imagery, movie, music, creative writing.	
Gen-AI was used in an assistive role to refine computer code and/or other output.	
Gen-AI was used in another assistive role (please describe):	
Gen-AI was fully used to complete a Category 3 assessment and use complied with the assessment instructions provided.	

By using this cover sheet and entering my student ID above, I declare that I have provided honest and accurate details regarding my use of Gen-AI within this assessment.

Abstract

In order to build immersive and captivating learning environments, extended reality technologies have become a crucial component of the evolving education industry. The design, construction, and assessment of an immersive virtual classroom utilizing browser-based WebXR technologies are covered in this study. The purpose of the application is to teach students the fundamentals of electricity, circuit components, current flow, and the distinctions between parallel and series circuits. Both desktop and virtual reality headset-based interactions are supported by the program, which is intended for usage with conventional web browsers. The design is based on research and analysis of existing relevant studies on WebXR-based virtual classrooms and immersive learning environments. A organized virtual area, 3D objects, and instructional media are all combined in the constructed program. To assess the application's efficacy and usefulness, a mixed-method assessment approach is suggested. With the use of suitable design concepts, the results demonstrate the potential of WebXR-based apps in developing captivating learning environments. This work contributes to the growing area of immersive virtual classrooms and Web-based XR-based learning environments in engineering education.

Table of Contents

1. Introduction 1

2. System implementation 2

2.1 System overview 2

2.2. Aims and objective 2

2.3. Technology Stack 3

2.4. System Architecture 4

2.5. Interactive Features 6

2.6. Deployment 7

3. Evolution Technique 16

3.1. Usability Testing 16

4. Learning Assessments 21

5. Technical Evaluation 21

6. Conclusion 21

7. References 22

Table of Figures

Figure 1: System architecture design 5

Figure 2 : System code -1 8

Figure 3 : System code -2 8

Figure 4 : System code -3 9

Figure 5: System code -4 9

Figure 6: System code -5 10

Figure 7: System code -6 10

Figure 8: System code -7 11

Figure 9: System code -8 11

Figure 10: System code -9 12

Figure 11: System code -10 12

Figure 12: System code -11 13

Figure 13: System code -12 13

Figure 14: System demonstration -1 14

Figure 15: System demonstration -2 14

Figure 16: System demonstration -3 15

Figure 17: Test -1 17

Figure 18: Test- 1.1 18

Figure 19: Test -2 19

Figure 20: Test -2.2 20

Figure 21: Test -2.3 20

Table of tables

Table 1 : Test -1 17

Table 2: Test -2 19

1. Introduction

In the field of education, virtual reality (VR), augmented reality (AR), and mixed reality (MR) technologies have begun to gain traction as ways to facilitate immersive learning. Immersion technologies have the potential to provide a greater sense of presence, allow students to interact with digital information in three dimensional space, and promote learning through exploring when compared to traditional teaching methodologies that use text or slide as teaching tools.

The project is based on the third scenario, which is immersive classroom and where a specific technical topic is taught in a browser-based virtual reality classroom. Virtual reality is a technology that let the users to experience and interact with a computer generated environment. In contrast to traditional screens, virtual reality covers the user's senses, particularly sight and sound. To detect user's brain into believing that they are in a virtual environment. It creates an immersive and realistic experience by simulating real or imagined settings using a combination of visual, audio and motion tracking techniques. [1]. In order to provide an immersive 3D experience, virtual reality technology combines hardware and software components. High resolution screens and optics for a realistic experience are among the hardware element. Sensors, monitor and user's motions, allowing the virtual environment to react realistically. Variety of input devices are used to make then user interactive device.

The reason behind choosing this topic is that it is directly involved with the learning material because the immersive environment can enhance student's engagement and comprehension and another is WebXR technology which offers a chance to create a learning tool that is easily accessible and doesn't require any complicated setup.

2. System implementation

2.1 System overview

The Greek term Systema, which denotes an ordered interaction between any group of elements to accomplish a shared goal or purpose, is where the word "system" originates [2]. The system is called WebXR-based virtual classroom which is for teaching fundamental concept. Users can experience and access the system through web browsers because it is designed as a browser based virtual reality environment.

2.2. Aims and objective

The aim of the project is to demonstrate the potential of a virtual reality classroom using a browser that provides efficient learning through a little applications. Objectives specify what must be achieved, how success will be evaluated, and what teams should prioritize in the event of trade-offs [3].

Here are some of the objective of this project is mentioned below.

- The primary objective of the Project is to design a structured immersive classroom.
- Implement interactive learning elements.
- Evaluate the effective of the system within an educational context.
- Understand how the system functions.
- Understand the reporting format.
- Research regarding technology stack.

2.3. Technology Stack

An arrangement of "things" kept one on top of the other is called a stack. Tech stacks are collections of technologies that may be assembled to create any kind of application. Tech stacks, often referred to as technology infrastructure or solutions stacks, are currently necessary for creating scalable and easily maintained online applications [3].

A list of all the technological services needed to create and operate a single application is termed a technology stack, technology infrastructure, or data ecosystem. The programming languages, frameworks, and tools a developer or product manager would require to communicate with the application are included in a tech stack. The tech stack indicates the overall strengths and weaknesses of the program because the majority of coding languages have well-known performance characteristics and limits [4].

The system was designed using the following technologies.

- 3D Scene Renderer
- JavaScript and HTML for interaction controls
- GitHub pages for deployment of the project.

A-Frame was chosen because of its ease of use and robust WebXR compatibility, which allow for the quick creation of dynamic 3D sceneries without the need for sophisticated graphics programming.

2.4. System Architecture

System architecture is the conceptual model that describes a system's behavior, structure and interaction between its many parts [5]. It is the process of making high-level choices regarding the system, such as the choice of software and hardware components, interface design, and the general architecture of the system.

The system architecture of this system illustrates that a Web-based XR (Extended Reality) applications, which enables a user to experience a virtual environment directly via a browser. The process commences at a user device that is computer that interacts with the application via a web browser such as chrome, Firefox or Microsoft edge. The main components of the system are built on WebXR and A-frame which facilitate immersive 3D experience in a browser based environment.

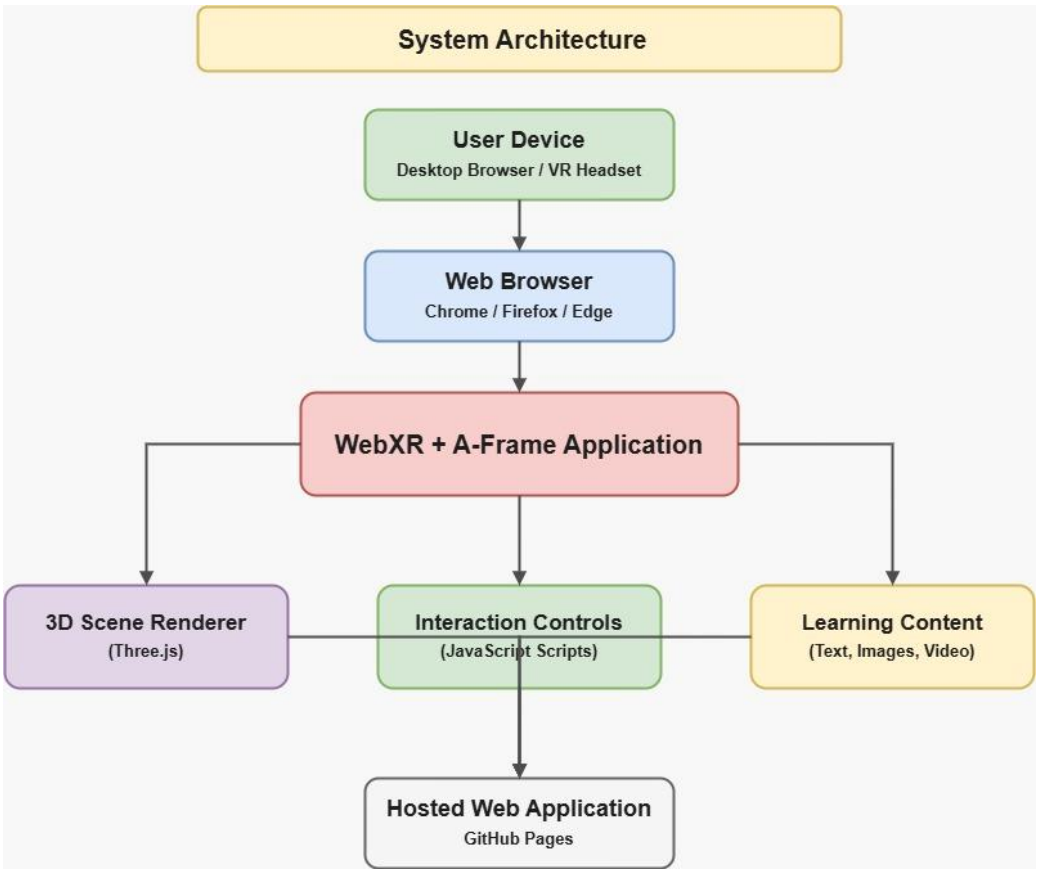

Figure 1: System architecture design

2.5. Interactive Features

The key features that the makes the system unique which includes are following features..

- Detect pictures and visuals
- Interactive 3D models

These elements boost engagement and promote active learning.

2.6. Deployment

The process of making a website or software program usable online or in a particular setting is called deployment. Deployment is transferring the finished code from a local system to a live server so that actual users may access and interact with it when development and testing are over [6]. It is more than last stage of development cycle. It's the point at which the program becomes operational and accessible to users [6]. GitHub pages was used to deploy the application, here is the link ***URL: https://github.com/AaratiMahato/Assignment-III.git***

GitHub is an online platform for social coding. An overview of GitHub's functionality, its use in professional and open-source development [7]. GitHub is an online platform that facilitates collaborative software development and hosts Git repositories. A few things to be aware of:

- Based on the Git system, it offers tools for project management, code review, version control, and problem tracking.
- GitHub is used by developers to create code and collaborate on open-source projects.

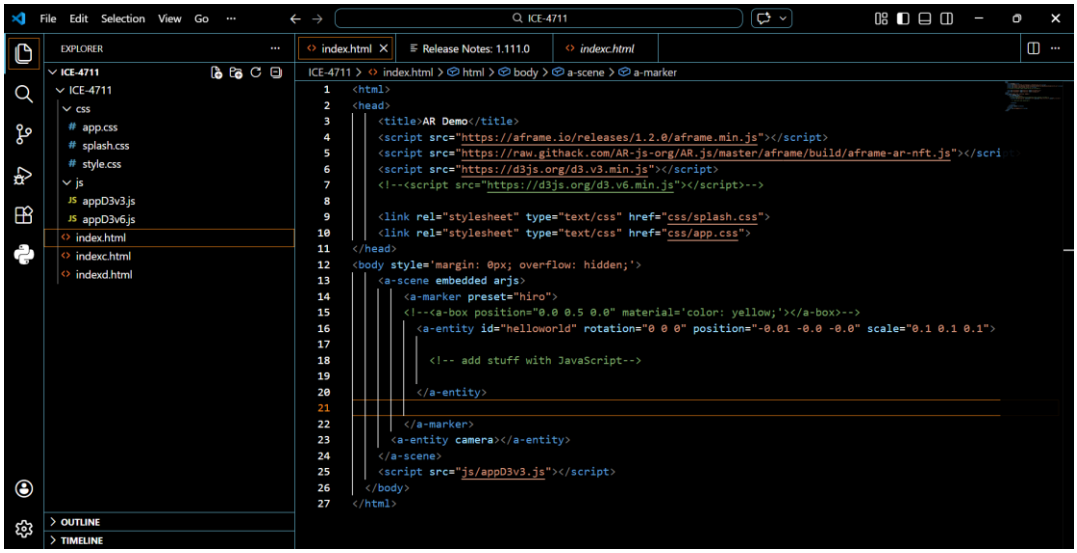

Figure 2 : System code -1

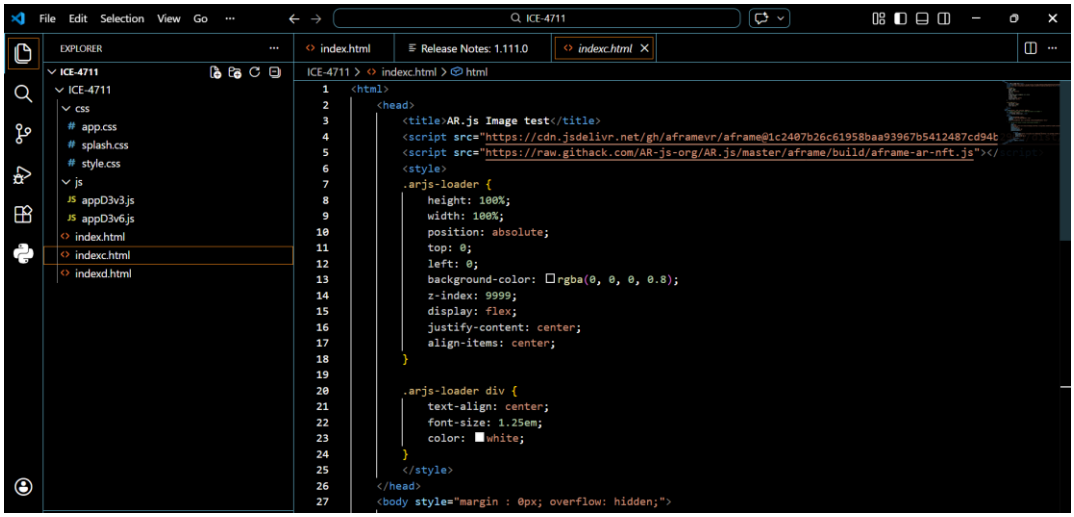

Figure 3 : System code -2

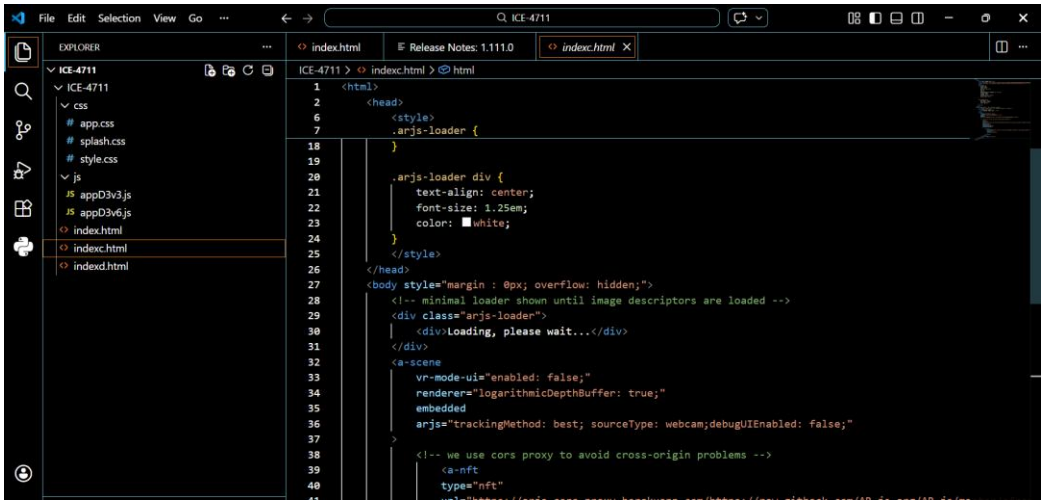

Figure 4 : System code -3

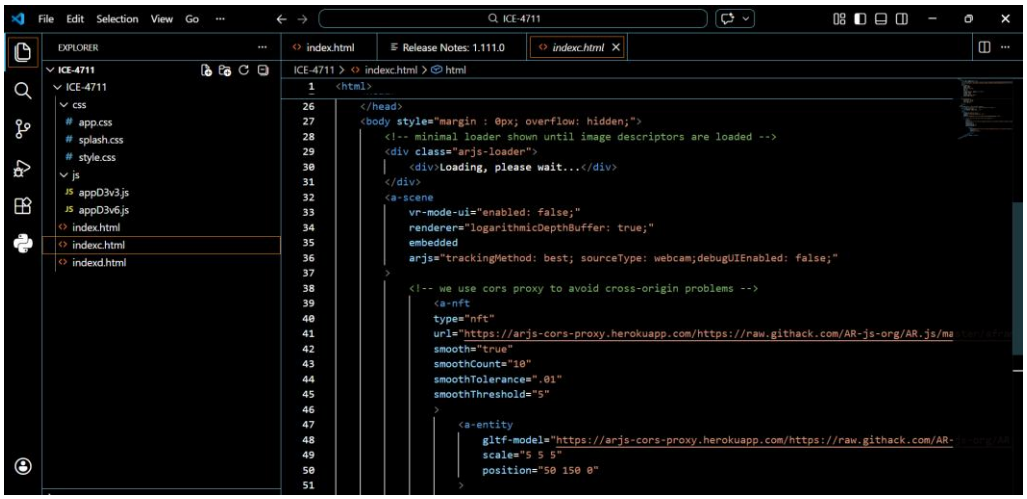

Figure 5: System code -4

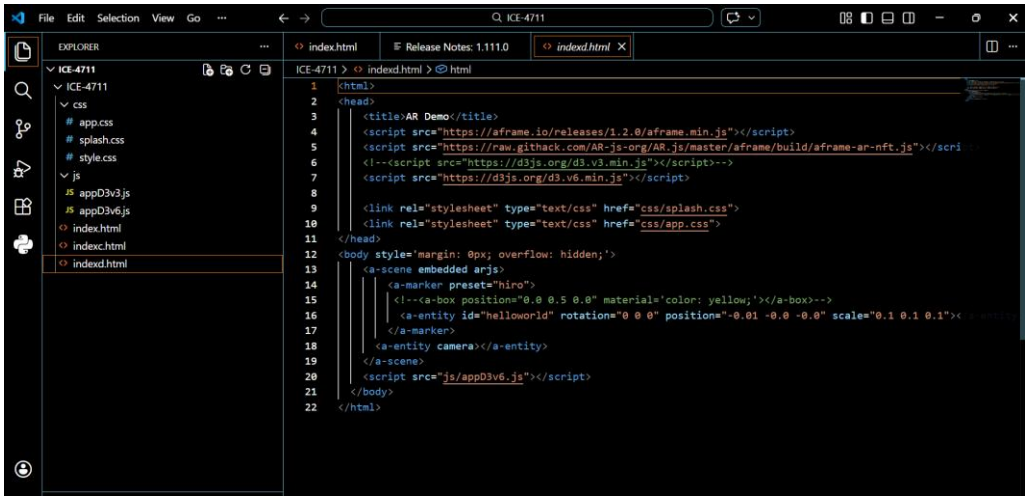

Figure 6: System code -5

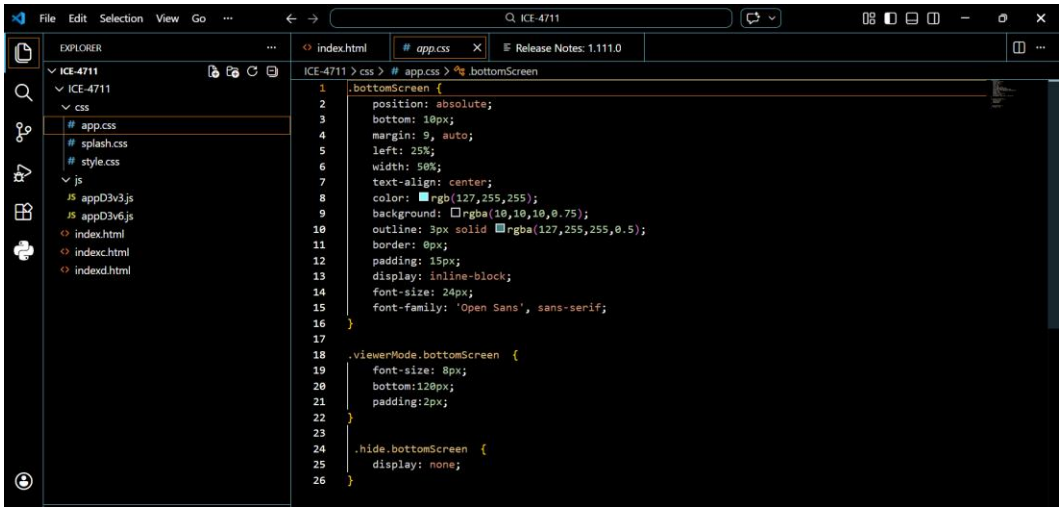

Figure 7: System code -6

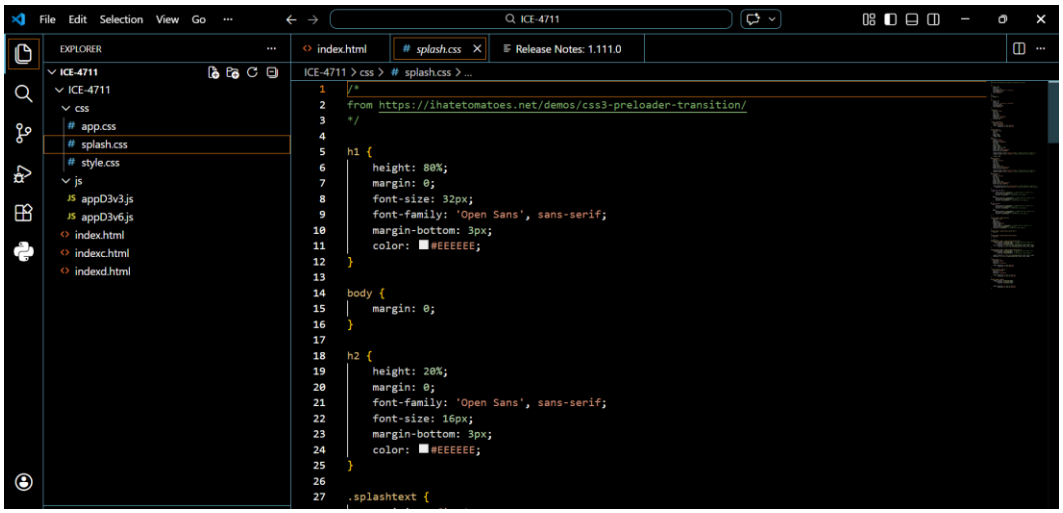

Figure 8: System code -7

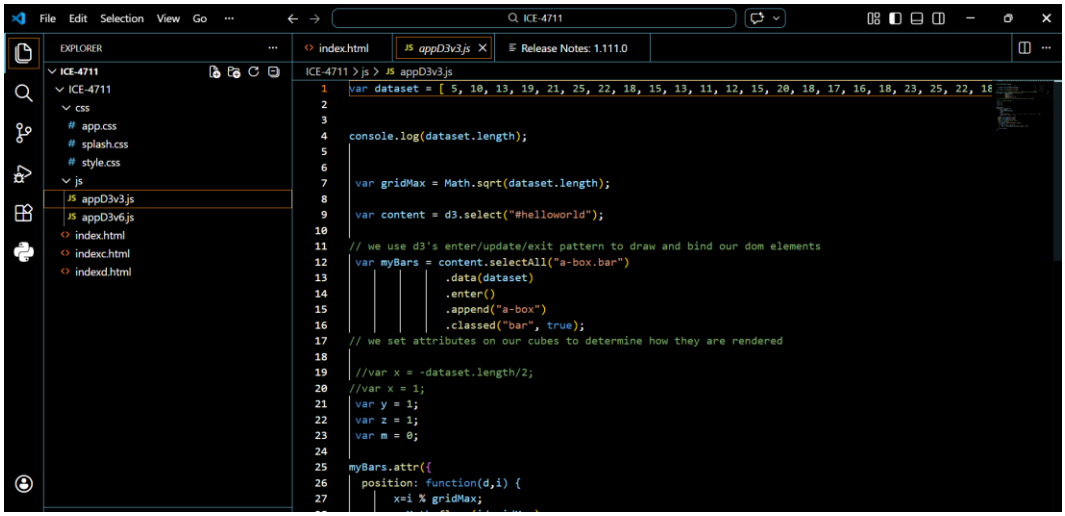

Figure 9: System code -8

Figure 10: System code -9

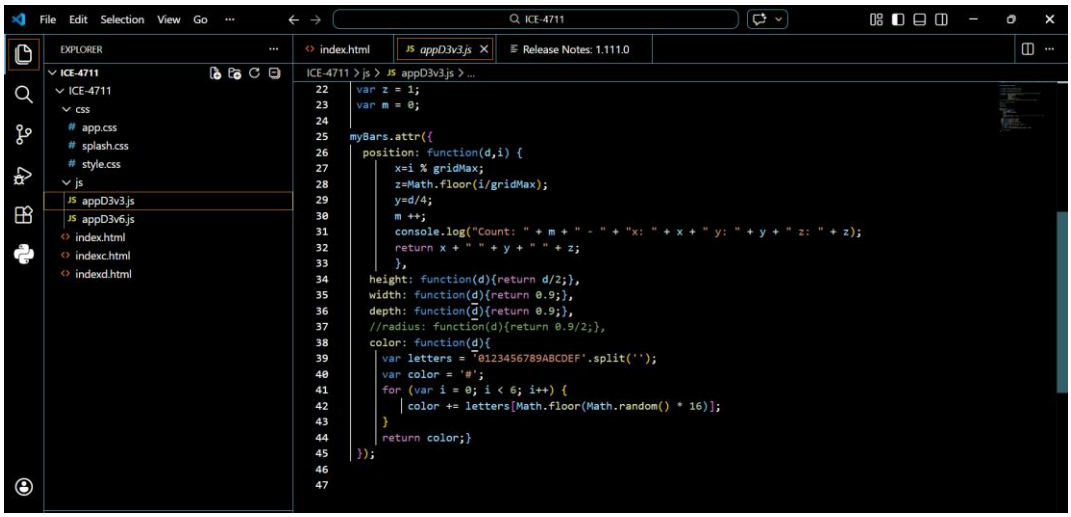

Figure 11: System code -10

Figure 12: System code -11

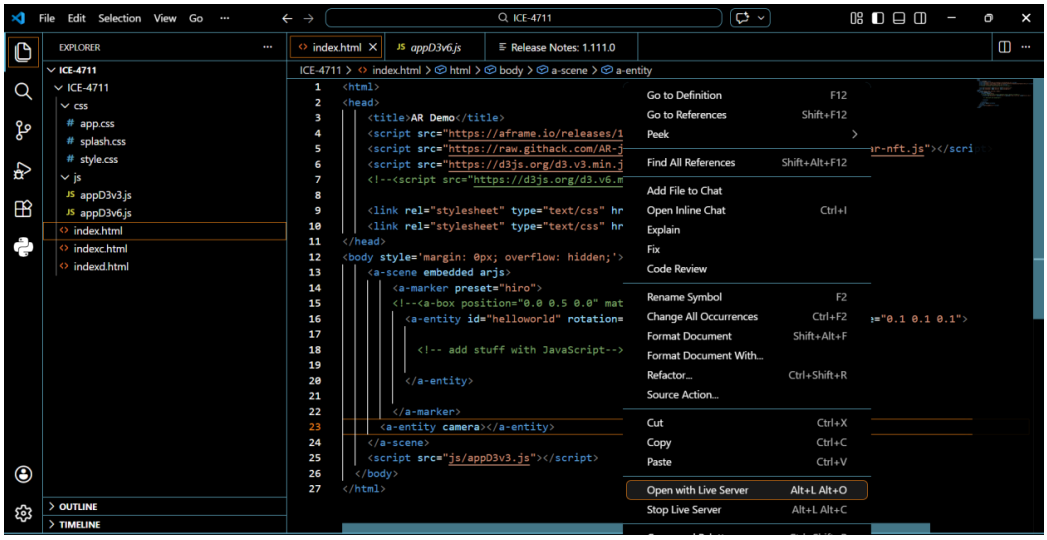

Figure 13: System code -12

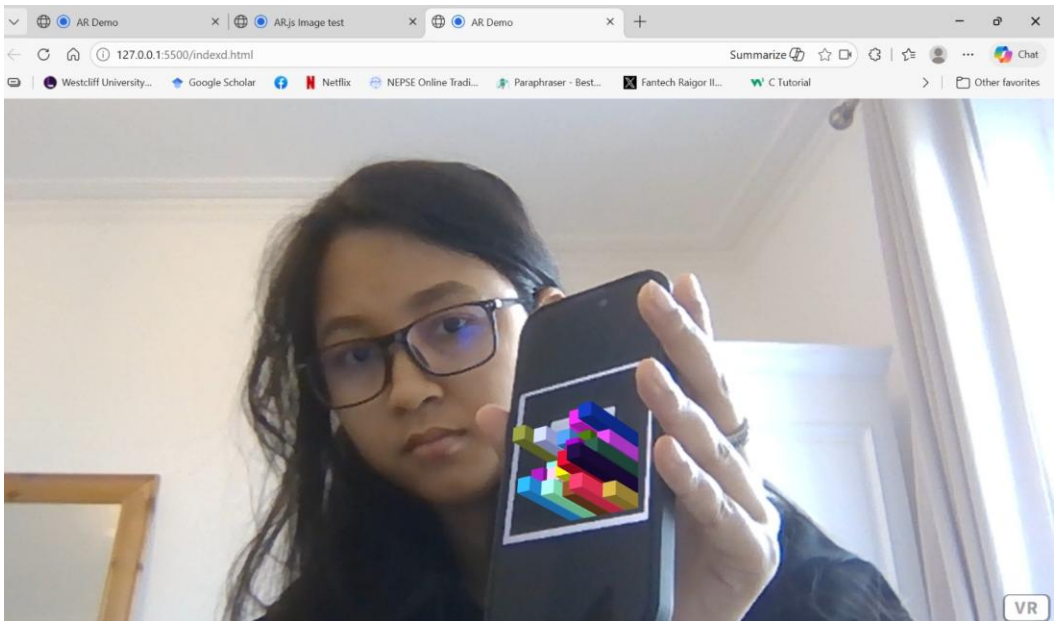

Figure 14: System demonstration -1

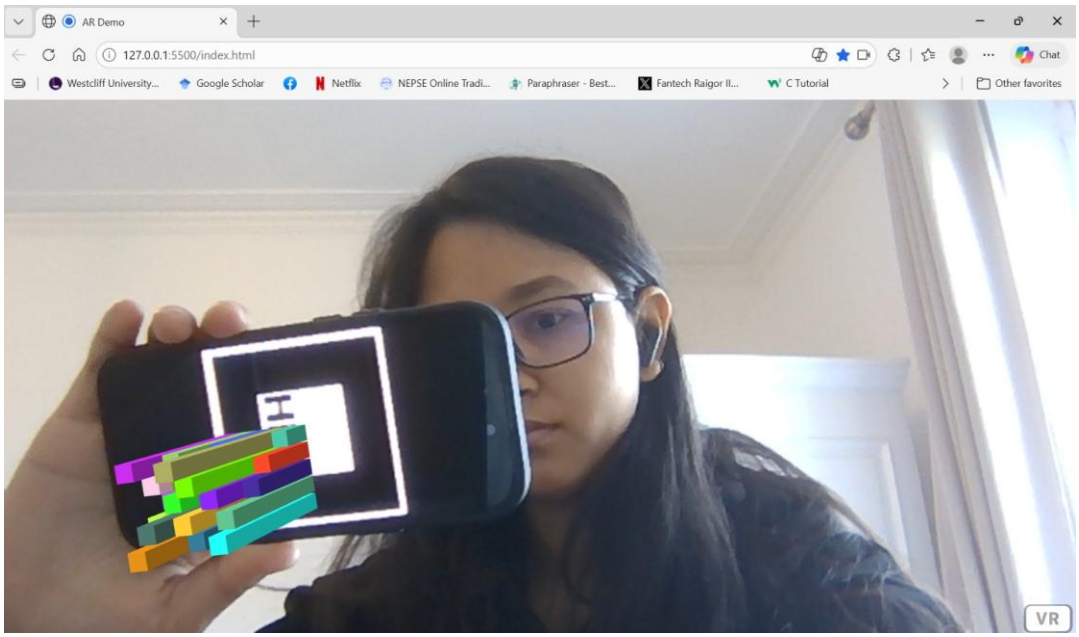

Figure 15: System demonstration -2

Figure 16: System demonstration -3

3. Evolution Technique

Evolution techniques, which are traditional variations of evolutionary algorithms, are widely employed to solve optimization problems heuristically, especially in continuous domains [8]. A methodical evaluation approach is suggested to gauge the success of the immersive classroom. This covers learning assessment, usability testing,

3.1. Usability Testing

The process of assessing and confirming that a software product or program operates accurately, securely, and effectively in accordance with its particular requirements is known as software testing [9]. It includes verification and validation procedures to find errors early and enhance the software quality and also stop performance or reliability problem in the future. Verification assures that the software is created in compliance with the requirements and design specifications. It provides a response to the query, “are we building the product correctly or not?” whereas validation guarantees that the program satisfies the requirement and expectations of the client that “are we building the right product?” is addressed [10].

Software defects can result in serious financial loss, system failure, and even fatalities. There are numerous historical examples that demonstrate the significance of software testing [10].

Test 1: To check the system is working or not.

Action	Expected result	Actual result	conclusion
First went to the code section in visual studio code and then clicked open with live server in index.html	The browser should open first followed by camera opening.	The browser and the camera was opened simultaneously.	The test was successfully carried out.

Table 1 : Test -1

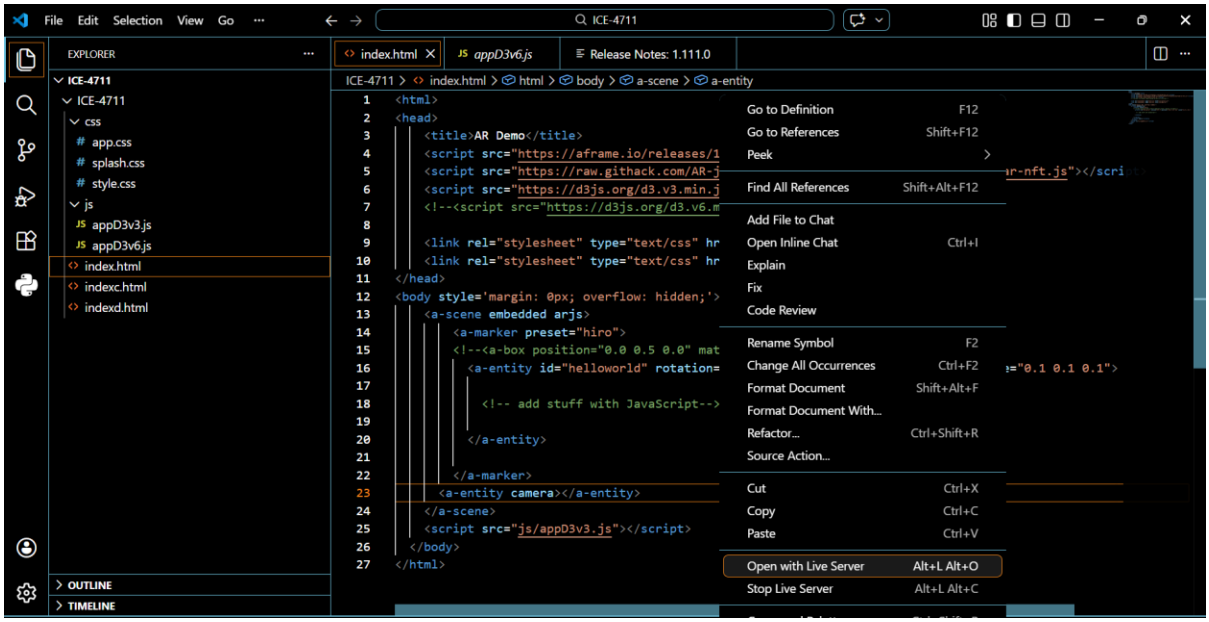

Figure 17: Test -1

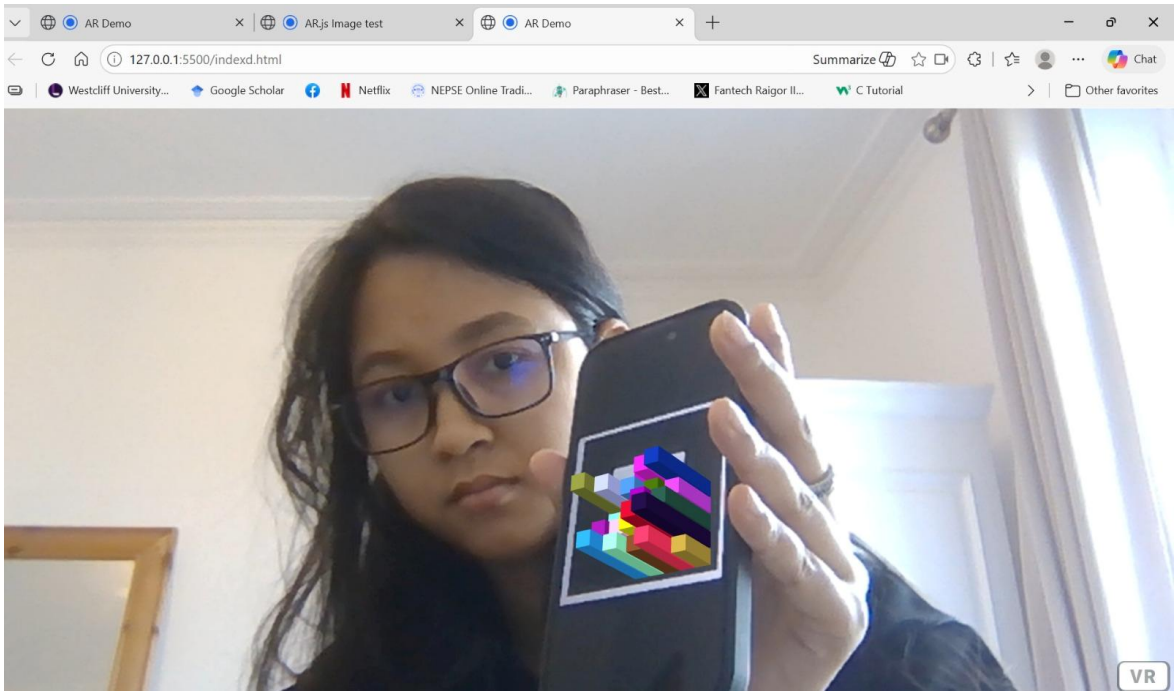

Figure 18: Test- 1.1

Test 2 : Testing Image

Action	Expected result	Actual result	conclusion
After opening camera the image named as Hero is showed towards the camera	3D module should be created.	3D module was created.	The test was successfully carried out.

Table 2: Test -2

Figure 19: Test -2

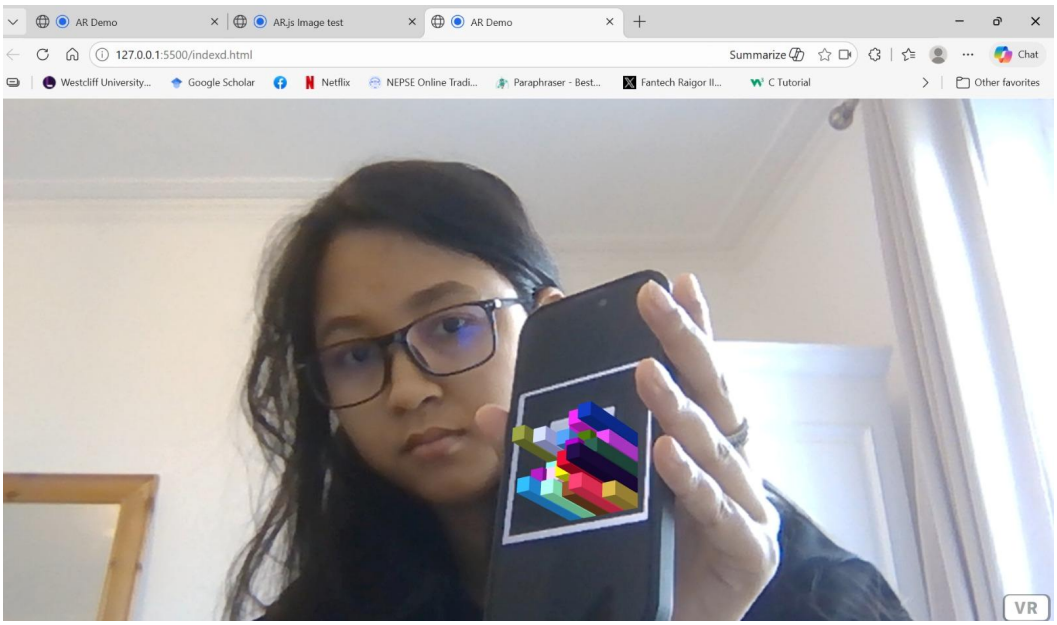

Figure 20: Test -2.2

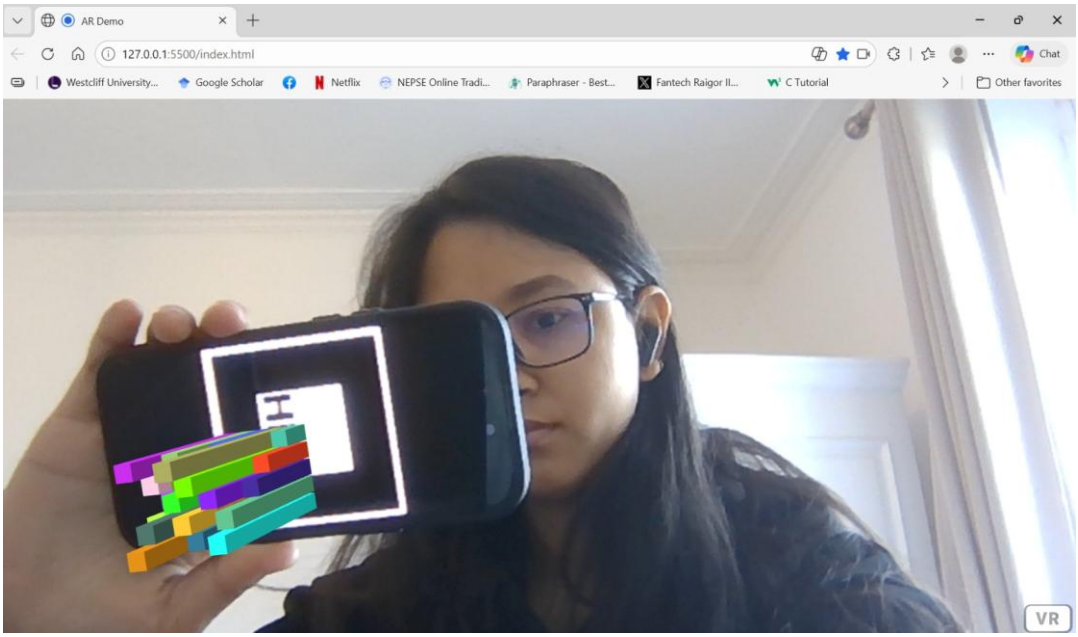

Figure 21: Test -2.3

4. Learning Assessments

To measure learning outcomes, a simple pre-test and post-test method can be employed. By answering questions both before and after using the system, participants may compare how much their knowledge has improved.

5. Technical Evaluation

One of the most important steps in the process of assessing the proposals that are presented in tenders is technical evaluation. Analyzing how well the submitted offer complies with the technical criteria and specifications listed in the tender papers is the main objective of this evaluation. To guarantee the selection of better offers that satisfy the project's criteria, technical review is typically carried out prior to price consideration [11]. Testing on many browsers and devices would guarantee performance and compatibility. This is especially crucial for WebXR apps.

6. Conclusion

The system demonstrates how virtual reality may enhance learning in ways that go beyond traditional methods. The WebXR application guarantees cross-platform accessibility, making it simple to use without requiring a laborious setup procedure. As long as there is a clear understanding of what is being learnt, the application shows that a straightforward XR application may be useful in teaching abstract engineering ideas. Additionally, the evaluation methodology shows the potential for learning and usability assessment. Despite the application's minimal interaction, it offers a solid platform for future improvements. The application makes a compelling argument for both the usage of WebXR to create a meaningful virtual world and the use of immersive technology in education.

7. References

[1] GeeksforGeeks, "Virtual Reality Introduction," 20 Feb 2026. [Online]. Available: https://www.geeksforgeeks.org/techtips/virtual-reality-introduction/.

[2] Tutorialspoint, "System Analysis and Design - Overview," 2026. [Online]. Available: https://www.tutorialspoint.com/system_analysis_and_design/system_analysis_and_design_overview.htm.

[3] MongoDB, Inc., "What is a Tech Stack and How Do They Work?," 2026. [Online]. Available: https://www.mongodb.com/resources/basics/technology-stack.

[4] Mixpanel, "What is a tech stack?," 2024. [Online]. Available: https://mixpanel.com/blog/what-is-a-technology-stack/.

[5] V. Tiwari, "Understanding System Architecture: A Beginner's Guide," 5 Feb 2025. [Online]. Available: https://medium.com/@vaibhavtiwari.945/understanding-system-architecture-a-beginners-guide-d2aad48f0007.

[6] GeeksforGeeks, "Deployment in Web Development: An Essential Guide," 23 Jul 2025. [Online]. Available: https://www.geeksforgeeks.org/websites-apps/deployment-in-web-development/.

[7] Coursera Inc, "What Is GitHub? Use Cases and FAQ," 16 Sep 2025. [Online]. Available: https://www.coursera.org/articles/what-is-github.

[8] M. Emmerich, "Evolution Strategies," 2018. [Online]. Available: https://www.researchgate.net/publication/332108895_Evolution_strategies.

[9] IBM, "What is software testing?," 2026. [Online]. Available: https://www.ibm.com/think/topics/software-testing.

[10] GeeksforGeeks, "What is Software Testing?," 9 Mar 2026. [Online]. Available: https://www.geeksforgeeks.org/software-testing/software-testing-basics/.

[11] TendersAlerts, "Technical Evaluation," 2026. [Online]. Available: https://tendersalerts.com/en/glossary/technical-evaluation.